

Notes: Card, Cardoso, and Kline (QJE, 2016)

Bargaining, Sorting, and the Gender Wage Gap: Quantifying the Impact of Firms on the Relative Pay of Women

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1 Big picture

- **Question.** How much of the gender wage gap is generated by firms, rather than only by worker characteristics? More specifically, do firms matter because women sort into lower-paying firms, because women receive smaller firm-specific pay premiums than men at the same firms, or both?
- **Setting.** The paper studies Portugal using matched employer-employee data from *Quadros de Pessoal* (QP) for 2002–2009, merged with firm financial statements from SABI. The design follows workers over time and uses mobility across firms to identify firm wage premiums.
- **Core challenge.** A firm-level contribution to the gender wage gap is hard to measure because workers with different permanent skill levels sort nonrandomly across firms. In addition, any estimate of a male-versus-female bargaining gap depends on how one normalizes firm effects, since firm premiums are only identified relative to a reference set of firms.
- **Main conceptual contribution.** The paper combines separate AKM-style worker-firm fixed-effect models for men and women with an Oaxaca-style decomposition. This lets the authors split the firm contribution to the gender wage gap into:
 - a *sorting effect*: women are underrepresented at firms that pay high premiums; and
 - a *bargaining effect*: women receive smaller premiums than men at the same firms.
- **Main hypotheses.**
 - If women are less likely to work at high-premium firms, the sorting channel widens the gender wage gap.
 - If women capture a smaller share of the surplus available at a given firm, the bargaining channel widens the gender wage gap.
 - If the bargaining interpretation is correct, the female-to-male premium ratio should show up both in cross-firm wage premiums and in within-firm wage responses to changes in profitability over time.
- **Main empirical design.** The authors:
 - estimate separate two-way fixed-effect wage models for male and female workers;
 - normalize the firm effects using firms with very low measured surplus;
 - decompose the cross-sectional firm component of the gender wage gap into sorting and bargaining parts; and
 - relate the estimated firm premiums to value added per worker and to changes in profitability over time.
- **Main baseline result.** In the dual-connected sample, the gender wage gap is 0.234 log points. Firm-specific pay premiums explain 0.049 log points of this gap, or about one-fifth of the total.
- **Main decomposition result.** Most of the firm contribution comes from sorting, but bargaining also matters. Depending on which gender distribution is used in the Oaxaca-style decomposition, the sorting effect explains about 0.035–0.047 log points, while the bargaining effect explains about 0.003–0.015 log points.

- **Main rent-sharing result.** The estimated firm premium is strongly related to excess log value added per worker. The slope is 0.156 for men and 0.137 for women, implying that women receive about 88%–90% of the firm-specific rent-sharing gains received by men.
- **Main dynamic result.** Among stayers, within-firm changes in excess value added per worker raise male wages with a coefficient of about 0.049 and female wages with a coefficient of about 0.045, implying a ratio of about 0.91. This lines up closely with the cross-firm evidence.
- **Economic takeaway.** Firms matter for the gender wage gap in two ways: women are less likely to work at high-rent firms, and even when they do, they capture a somewhat smaller share of the rents. The paper therefore shifts part of the gender wage-gap discussion away from pure worker-level explanations and toward firm wage-setting.

2 Data and measurement (key objects)

2.1 Institutional setting and why Portugal is a useful laboratory

- Portuguese women have relatively high labor-force participation, making the setting more comparable to the United States and Northern Europe than to economies where female employment is much lower.
- More than 90% of Portuguese women in private-sector jobs work full-time, so the analysis is less confounded by large cross-gender differences between full-time and part-time work.
- The gender wage gap in Portugal is close to the OECD average and not far from the gaps in the United States and the United Kingdom.
- During the sample period, about 90% of private-sector jobs were covered by sector-wide collective agreements. These contracts are nominally gender neutral, but firms still have substantial discretion over job assignment and wage cushions above contractual minima.
- Portugal also had a relatively high minimum wage, which could mechanically compress the lower tail of the gender wage gap. The paper addresses this concern explicitly later.

Interpretation. Portugal is useful because it combines broad female labor-force attachment, rich matched data, strong wage dispersion, and enough institutional structure to make the wage-setting environment interesting rather than purely competitive.

2.2 Data sources

- **Worker-firm data:** QP is an annual census of private-sector employees conducted each October by the Portuguese Ministry of Employment. Firms with at least one paid employee must report information for all workers.
- **Coverage:** The sample period is 2002–2009. The data contain roughly 4 million workers observed between one and eight times.
- **Financial data:** SABI contains firms' income statements and balance sheets. Coverage begins earlier, but is materially better after 2006.
- **Matching QP to SABI:** Because QP does not contain names or tax identifiers, the match relies on variables observed in both sources, including location, industry, firm creation date,

annual sales, and year-end shareholder equity.

Interpretation. QP identifies the worker side of wage determination, while SABI provides a direct measure of firm surplus. The merge is what allows the paper to connect estimated firm premiums to observable profitability.

2.3 Sample construction

- The main worker sample includes paid employees aged 19–65 with at least two years of potential labor-market experience and consistent employment histories.
- In the overall sample there are about 2.12 million men and 1.75 million women, generating about 9.07 million male person-year observations and 7.23 million female person-year observations.
- The authors estimate separate AKM models on the largest connected set for each gender. Roughly 91% of male person-years and 88% of female person-years are in those connected sets.
- For the male-female decomposition they focus on the *dual-connected* set of firms that lie in the connected sets for both genders.
- They also use narrower subsamples with matched value-added data, and later a stayer sample for the within-firm profitability analysis.

Interpretation. The connected-set restriction is not cosmetic. Without worker mobility linking firms, firm fixed effects are not identified. The dual-connected sample is the natural place to compare male and female firm premiums.

2.4 Wages, hours, and worker characteristics

- The wage outcome is the log of the real straight-time hourly wage.
- Hourly wages are built as:

$$\text{hourly wage} = \frac{\text{base salary} + \text{regular supplements}}{\text{normal hours of work}}.$$

- Regular supplements include recurring payments such as meal allowances. Overtime is excluded from the main hourly wage measure.
- Worker covariates include age, education, occupation, and location.

Interpretation. The hours information matters because it lets the authors study an hourly wage instead of a monthly earnings measure, which is important in a gender-wage-gap setting.

2.5 Firm surplus measures

- The main surplus proxy is mean log value added per worker, averaged over the years the firm is observed in SABI.
- Value added is the national-accounts concept: payments to labor and capital plus taxes, or equivalently revenues net of intermediate inputs under standard accounting assumptions.
- The alternative surplus proxy is mean log sales per worker.

Interpretation. Value added per worker is the main candidate for the bargaining surplus available at the firm. Sales per worker is noisier, but useful as a robustness check because it is observed more broadly and is less mechanically tied to labor costs.

2.6 Descriptive facts

- Women are slightly younger than men but more educated on average.
- In the overall sample women earn about 18% less per hour than men.
- Women work slightly fewer monthly hours than men, but the difference is small by international standards.
- Women are employed at somewhat larger firms than men on average.
- Gender segregation across firms is very strong: the average female worker is at a workplace that is about 70% female, while the average male worker is at a workplace that is only about 24% female.
- About 21% of men work at all-male firms and 19% of women work at all-female firms.
- Job mobility rates are remarkably similar for men and women. About three quarters of both groups hold only one job during the sample period, and survival rates of new jobs look similar across genders.

Interpretation. The descriptive evidence already hints that firms may matter through sorting because men and women work at very different firms. At the same time, the similarity in mobility rates makes it more plausible that differences in estimated firm premiums are not simply artifacts of different movement patterns.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The starting point is a rent-sharing model in which wages equal an outside option plus a gender-specific share of the surplus generated by the worker-firm match. The core parameter is the bargaining share ϕ_g for gender $g \in \{F, M\}$.

- If $\phi_F < \phi_M$, women receive a smaller fraction of the surplus at a given firm.
- If women are less likely to work at firms with large surplus, the sorting channel widens the wage gap.

Interpretation. The paper is not saying that all wage differences are rents. It is saying that the *firm-specific premium* part of wages can differ across gender both because of who works where and because of who gets how much surplus within a firm.

3.2 Rent-sharing model and two-way fixed effects

For worker i employed at firm $J(i, t)$ in year t , the log wage is:

$$w_{it} = a_{it} + \phi_{G(i)} S_{iJ(i,t),t}. \quad (1)$$

The surplus attached to the worker-firm match is decomposed as:

$$S_{iJ(i,t),t} = S_{J(i,t)} + \Delta_{J(i,t),t} + m_{iJ(i,t)}, \quad (2)$$

where:

- $S_{J(i,t)}$ is a time-invariant firm-wide surplus component,
- $\Delta_{J(i,t),t}$ is a time-varying firm-wide surplus component,
- $m_{iJ(i,t)}$ is a worker-firm-specific match component.

The outside option is written as:

$$a_{it} = \alpha_i + X'_{it} \beta_{G(i)} + \varepsilon_{it}, \quad (3)$$

where α_i is a worker fixed effect and X_{it} contains time-varying observables.

Combining these pieces gives the two-way fixed-effects wage equation:

$$w_{it} = \alpha_i + \psi_{J(i,t)}^{G(i)} + X'_{it} \beta_{G(i)} + r_{it}, \quad (4)$$

with

$$\psi_j^g = \phi_g S_j$$

and

$$r_{it} = \phi_{G(i)} \Delta_{J(i,t),t} + \phi_{G(i)} m_{iJ(i,t)} + \varepsilon_{it}.$$

Interpretation. The AKM firm effect ψ_j^g is interpreted as the gender-specific pay premium offered by firm j . Estimating the model separately by gender allows the premium paid by a firm to men and women to differ.

3.3 Exogeneity and mobility assumptions

The AKM estimates are obtained by OLS. For them to be unbiased, firm mobility has to be orthogonal to the time-varying residual component:

$$E[(r_{it} - \bar{r}_i)(D_{it}^j - \bar{D}_i^j) \mid G(i) = g] = 0 \quad \text{for all } j, \quad (5)$$

where D_{it}^j indicates whether worker i is at firm j in period t .

The paper discusses three threats:

- **Firm-wide shocks and mobility.** Workers may leave firms hit by negative shocks and join firms hit by positive shocks.
- **Comparative-advantage matching.** Movers may systematically improve their match component m_{ij} , making wage changes asymmetric.
- **Transitory worker shocks.** Workers with temporarily strong wage growth may move upward, while others move downward.

Interpretation. The descriptive mover graphs are designed to probe exactly these threats.

3.4 Normalization of the firm effects

Firm effects are only identified within connected sets and only up to an additive constant. To interpret them as rents, the paper needs a normalization.

Let S_j^o be an observed surplus measure, mainly mean log value added per worker. The identifying normalization is:

$$E[\psi_j^g \mid S_j^o < \tau] = 0, \quad (6)$$

where τ is a threshold below which firms are assumed to pay zero rents on average.

Operationally, the paper fits the piecewise-linear relation:

$$\hat{\psi}_j^g = \kappa_0^g + \kappa_g \max\{0, S_j^o - \tau\} + \nu_j^g, \quad (7)$$

and chooses τ to minimize the joint mean squared error for men and women.

Key estimate. The best-fitting threshold is $\hat{\tau} = 2.45$ in log value added per worker.

Interpretation. This is what converts “firm effects relative to an arbitrary reference firm” into “estimated rents relative to low-surplus firms.”

3.5 Oaxaca-style decomposition of the firm component

Let the average male premium be $E[\psi_J^M \mid \text{male}]$ and the average female premium be $E[\psi_J^F \mid \text{female}]$. Then:

$$\begin{aligned} E[\psi_J^M \mid \text{male}] - E[\psi_J^F \mid \text{female}] &= \underbrace{E[\psi_J^M - \psi_J^F \mid \text{male}]}_{\text{bargaining}} \\ &\quad + \underbrace{\left(E[\psi_J^F \mid \text{male}] - E[\psi_J^F \mid \text{female}] \right)}_{\text{sorting}} \end{aligned} \quad (8)$$

or, alternatively,

$$\begin{aligned} E[\psi_J^M \mid \text{male}] - E[\psi_J^F \mid \text{female}] &= \underbrace{E[\psi_J^M - \psi_J^F \mid \text{female}]}_{\text{bargaining}} \\ &\quad + \underbrace{\left(E[\psi_J^M \mid \text{male}] - E[\psi_J^M \mid \text{female}] \right)}_{\text{sorting}}. \end{aligned} \quad (9)$$

Interpretation.

- The *sorting effect* asks whether men and women are distributed differently across firms.
- The *bargaining effect* asks whether the same firm pays different premiums to men and women.
- Sorting is invariant to normalization.
- Bargaining is not invariant to normalization, so the paper interprets its baseline bargaining estimate as a lower bound.

3.6 Linking estimated firm effects to observable surplus

Define net surplus as:

$$NS_j = \max\{0, S_j^o - \tau\}. \quad (10)$$

The paper then estimates:

$$\psi_j^g = \beta_g NS_j + \omega_j^g. \quad (11)$$

Under the model,

$$\frac{\phi_F}{\phi_M} = \frac{\beta_F}{\beta_M}.$$

Interpretation. This is a direct rent-sharing regression. If the female slope is smaller than the male slope, women capture a smaller share of measurable rents.

3.7 Within-firm changes in wages over time

The paper also studies firm stayers. Let actual firm surplus be:

$$S_{jt} = \lambda NS_{jt} + \eta_{jt}. \quad (12)$$

For stayers at the same firm over time, wage changes satisfy:

$$\Delta w_{it} = \Delta X'_{it} \beta_{G(i)} + \delta_{G(i)} \Delta NS_{J(i)} + u_{it}, \quad (13)$$

with

$$\frac{\phi_F}{\phi_M} = \frac{\delta_F}{\delta_M}.$$

In practice, the paper estimates this in two steps:

- first, it constructs firm-level adjusted average wage changes separately for male and female stayers;
- second, it regresses the female firm-level change on the male firm-level change, instrumenting the male change with the change in measured surplus.

Interpretation. This dynamic design is valuable because it uses within-firm time variation rather than cross-firm wage levels. If both designs deliver the same bargaining ratio, the interpretation is much more convincing.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline AKM estimates

The baseline design does not use an external instrument. Identification comes from:

- worker mobility across firms,
- worker fixed effects, which absorb permanent ability,

- firm fixed effects, which capture systematic firm-level wage premiums,
- gender-specific estimation, which allows different premium schedules for men and women,
- and rich controls for age, education, and year effects.

Interpretation. The firm effect is identified by what happens to wages when comparable workers move across firms with different average wage policies.

4.2 Evidence from movers: why the additive model is plausible

The paper presents five descriptive facts from mover event studies:

- Workers moving to firms with higher-paid coworkers get wage gains; workers moving down get wage losses.
- There is no clear differential pretrend before moves to high- versus low-wage firms.
- Moves between firms with similarly paid coworkers produce little excess wage growth.
- Wage changes for opposite-direction moves are approximately symmetric.
- Women gain less than men from analogous upward moves and lose less from analogous downward moves.

Interpretation. The first four facts support the AKM framework and the exogenous-mobility approximation. The fifth fact is the paper's first reduced-form sign of a bargaining gap.

4.3 Why positive assortative matching does not invalidate the design

The paper finds positive correlations between estimated worker effects and estimated firm effects for both genders. That means higher-ability workers tend to sort to higher-premium firms.

This does *not* necessarily bias the AKM estimates because:

- the worker fixed effect absorbs permanent worker quality;
- the main identifying restriction concerns time-varying residuals and mobility, not time-invariant sorting.

Interpretation. Sorting on permanent skill is real and economically important, but it is not fatal for identification once worker fixed effects are included.

4.4 Normalization as a lower-bound strategy

The bargaining effect is sensitive to the normalization of firm effects. The paper's baseline assumption is that low-surplus firms pay zero rents on average.

This yields a lower-bound interpretation if:

- women at low-surplus firms do not receive larger rents than men at those firms.

Interpretation. The point is not that the zero-rent threshold is beyond doubt. The point is that it is a disciplined, data-driven normalization that makes the bargaining estimates interpretable and conservative.

4.5 Alternative normalization and alternative surplus proxies

The paper checks robustness in two important ways:

- It renormalizes the firm effects by assuming that the hotel and restaurant sector pays zero rents on average.
- It repeats the surplus analysis using sales per worker instead of value added per worker.

What happens?

- Sorting effects are unchanged, because they do not depend on normalization.
- Bargaining effects become larger under the hotel-and-restaurant normalization.
- The female-to-male rent-sharing ratio remains close to 0.88–0.90 when sales per worker is used.

Interpretation. The bargaining conclusion is not being manufactured by one arbitrary scaling choice.

4.6 Hours, amenities, and compensating differentials

A natural concern is that high-wage firms may simply require longer hours or less desirable schedules, which could generate compensating differentials rather than rent sharing.

The paper therefore relates estimated firm premiums to:

- mean regular hours,
- mean total hours including overtime,
- with and without industry controls,
- and with IV-style corrections for division bias using the other gender's hours.

Result. Once industry controls are added, the relationship between firm premiums and hours largely disappears.

Interpretation. The evidence does not support the idea that the firm premium structure is mainly compensation for longer hours, though the paper is careful not to rule out all unobserved job amenities.

4.7 Minimum-wage compression

Because Portugal's minimum wage is high, one concern is that lower-wage female workers may be artificially compressed upward, attenuating the measured bargaining gap.

The paper addresses this by re-estimating the models on workers aged 25 or older with at least a high-school education:

- only a tiny fraction of this subgroup is near the minimum wage,
- the bargaining effect is if anything larger,
- and the estimates are similar to those for university-educated workers in the full sample.

Interpretation. The paper’s main bargaining result is not just an artifact of minimum-wage compression at the bottom.

4.8 Where instruments actually enter

There is no external IV in the main cross-sectional decomposition. The one place where the paper uses an instrumental-variables logic is the stayer analysis:

- the adjusted average change in female wages at a firm is regressed on the adjusted average male change;
- the change in measured surplus is used as an instrument for the male change.

Interpretation. This step is not the main source of the paper’s baseline result. It is a cross-check that the female-to-male bargaining ratio estimated from within-firm time variation resembles the ratio estimated from between-firm wage premiums.

4.9 What the paper can and cannot distinguish

- The paper is strong on showing that firms matter through both sorting and differential surplus sharing.
- It is weaker on the deeper source of the bargaining gap. The gap could reflect differences in negotiation, monopsony, discrimination, turnover elasticities, or other frictions.
- The decomposition also cannot recover the counterfactual wages women would receive at all-male firms or men would receive at all-female firms, which is why the main analysis focuses on mixed-gender connected sets.

Interpretation. This is a paper about *where* the gender wage gap comes from in the wage structure, not a final theory of *why* women and men bargain differently.

4.10 Relation to the literature

The paper ties together three strands of work:

- the AKM literature on worker and firm fixed effects,
- the rent-sharing literature linking wages to firm surplus,
- and the gender wage-gap literature on workplace segregation and unequal treatment within firms.

Its main advance is to show how firm-specific premiums can be decomposed into sorting and bargaining components in a matched employer-employee framework.

5 Tables and figures

5.1 Table I: sample construction and descriptive structure

Read:

- In the overall sample, mean log real hourly wages are 1.59 for men and 1.41 for women.

TABLE I
DESCRIPTIVE STATISTICS FOR VARIOUS SAMPLES OF EMPLOYEES IN QUADROS DE PESSOAL (QP), 2002-2009

	(1)	(2)	(3) Connected Sets of Workers/Firms				(7)	(8)
	Overall Analysis Sample		All		Dual-Connected		Analysis Sample with Value Added Data	
	Males	Females	Males	Females	Males	Females	Males	Females
Age:								
Mean age	38.1	36.9	38.0	36.5	38.0	36.4	37.9	36.3
Fraction ≤ 30 years old	0.30	0.33	0.30	0.34	0.30	0.34	0.31	0.35
Fraction ≥ 50 years old	0.19	0.14	0.18	0.13	0.19	0.13	0.18	0.13
Education:								
Mean years schooling	8.0	8.8	8.0	8.9	8.6	9.1	7.8	8.6
Fraction with high school	0.18	0.23	0.18	0.23	0.21	0.24	0.17	0.24
Fraction with degree	0.09	0.13	0.09	0.14	0.11	0.15	0.07	0.11
Mean log real hourly wage	1.59	1.41	1.62	1.43	1.71	1.48	1.54	1.36
(std. dev.)	(0.55)	(0.50)	(0.55)	(0.51)	(0.58)	(0.53)	(0.48)	(0.43)
Mean monthly hours	162.6	158.0	162.5	157.9	162.8	157.1	164.1	159.9
(std. dev.)	(24.7)	(30.1)	(24.8)	(29.9)	(24.0)	(30.5)	(23.6)	(29.4)
Fraction in Lisbon	0.35	0.35	0.36	0.37	0.42	0.40	0.32	0.34
Fraction in Oporto	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.14
Mean firm size (no. employees)	730	868	804	978	1,091	1,230	500	888
Fraction females at firm	0.24	0.70	0.24	0.70	0.30	0.64	0.24	0.67
Mean log VA/worker							3.05	2.88
Number person-year obs.	9,070,492	7,226,310	8,225,752	6,334,039	6,012,521	5,012,736	5,786,108	4,204,623
Number of persons	2,119,687	1,747,492	1,889,366	1,505,517	1,450,288	1,247,503	1,441,608	1,082,052
Number of firms	349,692	336,239	216,459	185,086	84,720	84,720	153,994	141,887

Notes: Overall analysis sample in columns (1)-(2) includes paid workers age 19-65 with potential experience ≥ 2 , and consistent employment histories. See Section A of Online Appendix. Wages are measured in real (2009 = 100) euros per hour. Value added (VA) is measured in thousands of real euros per year. All statistics are calculated across person-year observations. See text for definitions of connected and dual connected sets.

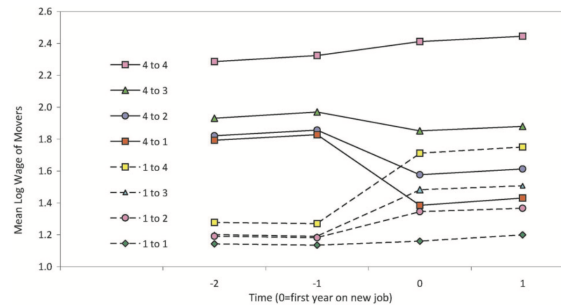


FIGURE I
Mean Log Wages of Male Job Changers, Classified by Quartile of Mean Co-Worker Wage at Origin and Destination Firm

Figure shows mean wages of male workers at mixed-gender firms who changed jobs in 2004-2007 and held the preceding job for two or more years, and the new job for two or more years. Each job is classified into quartiles based on mean log wage of coworkers of both genders in the last year of the old job (for origin firm) and in the first year on the new job (for the destination firm). See text for additional details.

- Women have more schooling on average, but still earn less and work slightly fewer monthly hours.
- Women work at larger firms on average.
- The average female share at women's firms is 0.70, versus 0.24 at men's firms.
- In the dual-connected sample, the gender wage gap rises to 0.234 log points.

Economic message. Firm segregation is quantitatively large, and once single-gender firms are removed the remaining mixed-gender comparison sample still has a large gender wage gap.

5.2 Figures I-III: mover evidence on firm premiums and gender differences

Read:

- Figures I and II show wage paths for men and women who move between quartiles of coworker wages.
- Moving to higher-wage firms produces large gains; moving to lower-wage firms produces large losses.
- Pre-move trends are similar across destination groups.
- Opposite-direction moves produce roughly symmetric gains and losses.
- Figure III plots adjusted wage changes for women against those for men across origin-destination cells and yields a slope of about 0.77.

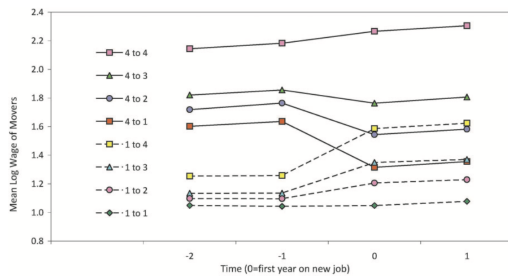


FIGURE II

Mean Wages of Female Job Changers, Classified by Quartile of Mean Co-Worker Wage at Origin and Destination Firm

Figure shows mean wages of female workers at mixed gender firms who changed jobs in 2004-2007 and held the preceding job for two or more years, and the new job for two or more years. Each job is classified into quartiles based on mean log wage of coworkers of both genders in the last year of the old job (for origin firm) and in the first year on the new job (for the destination firm). See text for additional details.

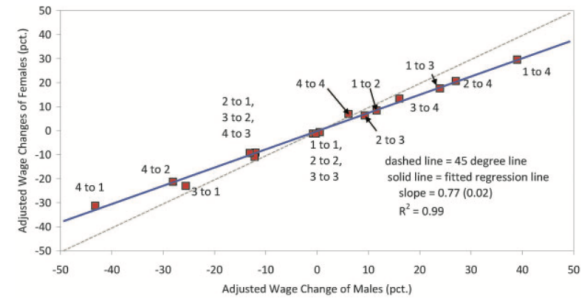


FIGURE III

Comparison of Adjusted Wage Changes of Male and Female Job Movers by Quartile of Coworker Wages at Origin and Destination Firms

Points represent regression adjusted mean log wage changes of male and female job movers in different origin/destination quartiles of mean coworker wages. For example "4 to 1" point shows mean wage changes for men and women who move from 4th quartile of coworker wages to 1st quartile. Fitted line is estimated by OLS to 16 points.

TABLE II
SUMMARY OF ESTIMATED TWO-WAY FIXED EFFECTS MODELS FOR MALE AND FEMALE WORKERS

	(1) All Males	(2) All Females
Standard deviation of log wages	0.554	0.513
Number of person-year observations	8,225,752	6,334,039
Summary of parameter estimates:		
Number person effects	1,889,366	1,505,517
Number firm effects	216,459	185,086
Std. dev. of person effects (across person-yr obs.)	0.420	0.400
Std. dev. of firm effects (across person-yr obs.)	0.247	0.213
Std. dev. of Xb (across person-yr obs.)	0.069	0.059
Correlation of person/firm effects	0.167	0.152
RMSE of model	0.143	0.125
Adjusted R-squared of model	0.934	0.940
Correlation of estimated male/female firm effects	0.590	
Comparison job-match effects model:		
Number of job-match effects	2,689,648	2,087,590
RMSE of match-effects model	0.128	0.113
Adjusted R-squared of match-effects model	0.946	0.951
Standard deviation of job match effect	0.062	0.054
Inequality decomposition of two-way fixed effects model:		
Share of variance of log wages due to:		
Person effects	57.6	61.0
Firm effects	19.9	17.2
Covariance of person and firm effects	11.4	9.9
Xb and associated covariances	6.2	7.5
Residual	4.9	4.4

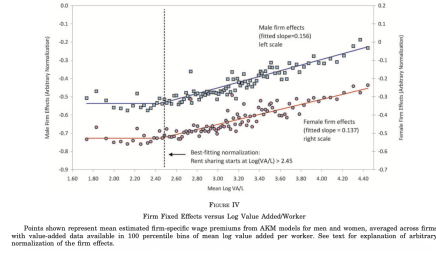
Notes. See text. Models include dummies for individual workers and individual firms, year dummies interacted with education dummies, and quadratic and cubic terms in age interacted with education dummies (total of 44 parameters). Comparison job-match effects models include dummies for each worker-firm job match as well as other covariates in basic model. Samples include only observations in largest connected set.

Economic message. The mover graphs support the existence of firm-specific wage premiums and show, in a very transparent reduced-form way, that women gain less than men from the same type of upward firm-to-firm mobility.

5.3 Table II: the two-way fixed-effects models

Read:

- The standard deviation of worker effects is 0.420 for men and 0.400 for women.
- The standard deviation of firm effects is 0.247 for men and 0.213 for women.
- The correlation between worker and firm effects is positive for both genders: 0.167 for men and 0.152 for women.
- The correlation between estimated male and female firm effects is 0.590.
- Firm effects explain about 19.9% of male wage variance and 17.2% of female wage variance.
- The covariance between person and firm effects explains another 11.4% of male wage variance



and 9.9% of female wage variance.

- A generalized job-match model improves fit only modestly; the implied standard deviation of job-match effects is 0.062 for men and 0.054 for women.

Economic message. Firms are an important source of wage inequality, but the shared firm premium component is much bigger than the match-specific component. This makes firm-level decomposition economically meaningful.

5.4 Figure IV: the normalization strategy

Read:

- Figure IV plots estimated firm effects against mean log value added per worker.
- The relationship is flat at the very bottom and then becomes approximately linear.
- The best-fitting kink occurs at log value added per worker of 2.45.
- The fitted slopes above the threshold are 0.156 for men and 0.137 for women.

Economic message. The paper’s zero-rent normalization is not imposed blindly. It is anchored in a visible kink in the relationship between firm effects and measured surplus.

5.5 Figure V and Table V: women receive about 90% of male rent-sharing gains

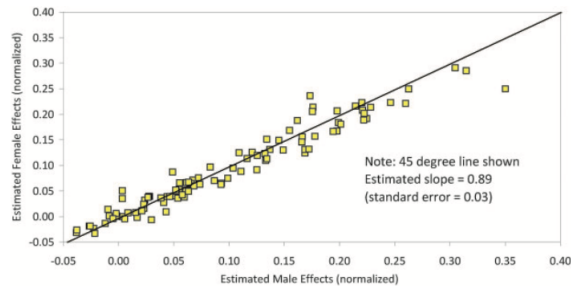


FIGURE V
Estimated Firm Effects for Female and Male Workers: Firm Groups Based on Mean Log Value Added per Worker

TABLE V
ESTIMATED RELATIONSHIP BETWEEN GENDER-SPECIFIC FIRM EFFECTS AND MEASURES OF SURPLUS PER WORKER

	(1)	(2)	(3)	(4)
		Regressions of Firm Effects on Measure of Surplus:		
	Number of Firms	Male Firm Effects	Female Firm Effects	Ratio: Column (3) / Column (2)
Surplus measure:				
1. Excess mean log value added per worker	47,477	0.156 (0.006)	0.137 (0.006)	0.879 (0.031)
2. Mean log sales per worker	75,163	0.072 (0.005)	0.064 (0.004)	0.897 (0.036)
3. Excess mean log sales per worker	75,163	0.092 (0.006)	0.081 (0.006)	0.883 (0.038)

Notes. Columns (2)–(3) report coefficients of surplus measure indicated in row heading in regression models in which the dependent variables are the estimated firm effects for males or females. All specifications include a constant and are estimated at the firm level, weighting by the total number of male and female workers at the firm. Ratios in column (4) are estimated by instrumental variables, treating the firm effect in female wages as the dependent variable, the firm effect in male wages as the endogenous explanatory variable, and the surplus measure as the instrument. Standard errors, clustered by firm, are in parentheses.

TABLE III
CONTRIBUTION OF FIRM-SPECIFIC PAY PREMIUMS TO THE GENDER WAGE GAP AT DUAL-CONNECTED FIRMS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Means of Firm Premiums			Decompositions of Contribution of Firm Component				
	Gender Wage Gap	Male Premium among Men	Female Premium among Women	Total Contribution of Firm Components	Sorting		Bargaining	
					Using Male Effects	Using Female Effects	Using Male Distribution	Using Female Distribution
All	0.234	0.148	0.099	0.049 (21.2)	0.035 (14.9)	0.047 (19.9)	0.003 (1.2)	0.015 (6.3)
By age group:								
Up to age 30	0.099	0.114	0.087	0.028 (28.1)	0.019 (18.9)	0.029 (29.3)	-0.001 (-1.2)	0.009 (9.3)
Ages 31-40	0.228	0.156	0.111	0.045 (19.7)	0.029 (12.6)	0.040 (17.8)	0.004 (1.9)	0.016 (7.0)
Over age 40	0.336	0.169	0.099	0.069 (20.6)	0.050 (15.0)	0.064 (19.1)	0.005 (1.5)	0.019 (5.6)
By education group:								
< High school	0.286	0.115	0.055	0.059 (20.8)	0.045 (15.6)	0.061 (21.4)	-0.002 (-0.6)	0.015 (5.2)
High school	0.262	0.198	0.137	0.061 (23.3)	0.051 (19.6)	0.051 (19.5)	0.010 (3.8)	0.010 (3.7)
University	0.291	0.259	0.213	0.047 (16.1)	0.025 (8.7)	0.029 (9.9)	0.018 (6.2)	0.022 (7.4)

Notes: Sample includes male and female workers in dual connected set (Table I, columns (2)-(5)). Entry in column (1) is the difference in mean log wages of males and females, estimated over all workers in the subset of the dual-connected set indicated by the row heading. Estimated firm effects are from models described in columns (1) and (2) of Table II. Entry in column (4) is the total contribution of firm-specific wage premiums to the gender wage gap reported in column (1). Entries in columns (5)-(8) are the contributions of sorting effect and bargaining effect to gender wage gap, calculated using method described in the text. Entries in parentheses represent the percent of the overall male-female wage gap (in column (1)) that is explained by the source described in column heading.

Read:

- Figure V groups firms by value added per worker and plots female firm effects against male firm effects.
- The grouped-data slope is about 0.89.
- Table V confirms this in direct rent-sharing regressions:
 - with excess log value added per worker, the slopes are 0.156 for men and 0.137 for women, ratio 0.879;
 - with mean log sales per worker, the slopes are 0.072 and 0.064, ratio 0.897;
 - with excess mean log sales per worker, the slopes are 0.092 and 0.081, ratio 0.883.

Economic message. The female firm premium rises with firm surplus almost one-for-one with the male premium, but not quite. The shortfall is small in percentage terms, yet economically meaningful because it affects the entire cross-firm premium schedule.

5.6 Table III, Figure VI, and Table IV: how the gender gap is decomposed

Read:

- For all workers in the dual-connected sample, the gender wage gap is 0.234 log points.
- Mean firm premium among men is 0.148; mean firm premium among women is 0.099.
- The total firm contribution is therefore 0.049 log points, about 21% of the total gap.
- Sorting explains about 0.035–0.047 log points, depending on whether one uses male or female firm effects in the decomposition.
- Bargaining explains about 0.003–0.015 log points in the baseline normalization.
- Figure VI shows that the firm contribution grows with age, mainly because sorting rises over the life cycle.
- Table III also shows that sorting is more important for low- and middle-educated workers, while bargaining is relatively larger for university-educated workers.
- Table IV shows substantial heterogeneity by occupation. The firm component is especially

important for technicians, clerks, craft workers, and elementary occupations; bargaining is relatively more important for managers and professionals.

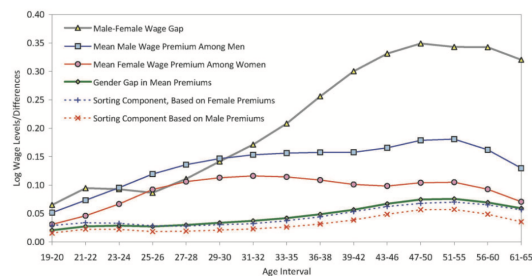


FIGURE VI
Evolution of Gender Wage Gap and Its Components over the Life Cycle
Figure shows unadjusted male-female wage gap, means of firm-specific wage premiums earned by men and women, and the difference in mean premiums, which is the total contribution of firm-specific wage components to the gender wage gap. Dashed lines show the effect of differential sorting of males and females to specific firms, evaluated using male and female firm-specific wage premiums.

TABLE IV

CONTRIBUTION OF FIRM-LEVEL PAY COMPONENTS TO GENDER WAGE GAP, BY MODAL OCCUPATION

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
		Means of Firm Premiums:			Decompositions of Contribution of Firm Component:			
					Sorting		Bargaining	
	Gender Wage Gap	Male Premium among Men	Female Premium among Women	Total Contribution of Firm Components	Using Male Effects	Using Female Effects	Using Male Distribution	Using Female Distribution
All	0.234	0.148	0.099	0.049 (21.2)	0.035 (14.9)	0.047 (19.9)	0.003 (1.2)	0.015 (6.3)
By modal occupation:								
Managers (69% male)	0.238	0.219	0.210	0.010 (4.0)	-0.005 (-2.1)	-0.003 (-1.4)	0.013 (5.4)	0.014 (6.1)
Professionals (51% male)	0.153	0.253	0.221	0.032 (20.9)	0.007 (4.3)	0.016 (10.3)	0.016 (10.6)	0.025 (16.6)
Technicians (65% male)	0.157	0.238	0.179	0.059 (37.5)	0.040 (25.4)	0.039 (24.5)	0.013 (5.3)	0.019 (12.1)
Clerks (41% male)	0.128	0.217	0.161	0.056 (43.8)	0.044 (34.4)	0.047 (36.6)	0.009 (7.1)	0.012 (9.4)
Services (32% male)	0.145	0.055	0.046	0.008 (5.8)	0.009 (6.0)	0.020 (13.9)	-0.012 (-8.0)	0.000 (0.2)
Craft (68% male)	0.389	0.088	0.016	0.072 (18.6)	0.041 (10.5)	0.078 (20.1)	-0.006 (-1.5)	0.031 (8.1)
Operatives (75% male)	0.303	0.150	0.104	0.045 (15.0)	0.032 (10.6)	0.036 (11.9)	0.009 (3.1)	0.013 (4.4)
Elementary (47% male)	0.196	0.109	0.055	0.054 (27.6)	0.033 (16.6)	0.066 (33.7)	-0.012 (-6.1)	0.022 (11.0)

Notes: see notes to Table III. Workers are classified into their most common occupation during years they are observed. Farm and fishing workers are included with operatives.

Economic message. The central decomposition result is that firms matter a lot, mostly through sorting overall, but with a bargaining component that becomes more prominent in high-skill segments of the labor market.

5.7 Table VI: within-firm changes in profitability and wages

Read:

- For stayers between 2006 and 2009, the rent-sharing coefficients using changes in excess log value added per worker are 0.049 for men and 0.045 for women, implying a ratio of 0.911.
- Without Winsorizing the profitability change, the coefficients fall to 0.035 and 0.031, but the ratio remains 0.894.
- Using changes in excess log sales per worker yields coefficients of 0.021 and 0.018, ratio 0.876.

Economic message. The time-series evidence from firm stayers points to essentially the same female-to-male bargaining ratio as the cross-sectional firm-premium evidence. That is one of the most persuasive parts of the paper.

6 Short summary

- The paper asks how much of the gender wage gap is generated by firms through sorting and differential bargaining.
- Using Portuguese matched employer-employee data, it estimates separate worker-firm fixed-effect models for men and women.
- Women are less likely to work at firms that pay high premiums to either gender.
- Women also receive smaller firm-specific premiums than men at the same firms.
- In the dual-connected sample, firm-specific premiums explain about 21% of the overall gender wage gap.
- Most of that firm contribution comes from sorting, but bargaining is nontrivial and more important for highly educated workers.

TABLE VI
EFFECTS OF CHANGES IN MEASURED SURPLUS PER WORKER ON THE CHANGE IN WAGES
OF STAYERS

Surplus measure and sample:	(1)	(2)	(3)	(4)
	Number of Firms	Estimated Rent Sharing Coefficients:		Ratio: Column (3) / Column (2)
		Male Stayers	Female Stayers	
1. Excess log value added per worker (Winsorized at +/- 0.50). Sample = stayers at firms with value added data, 2006–9	33,104	0.049 (0.007)	0.045 (0.008)	0.911 (0.086)
2. Excess log value added per worker (not Winsorized). Sample = stayers at firms with value added data, 2006–9	33,104	0.035 (0.006)	0.031 (0.006)	0.894 (0.091)
3. Excess log sales per worker (Winsorized at +/- 0.50). Sample = stayers at firms with sales data, 2005–8	44,266	0.021 (0.006)	0.018 (0.005)	0.876 (0.182)

Notes. Dependent variables are average change in wages of male or female workers at a firm (regression-adjusted for quadratic in age). Table entries are coefficients of the measured change in surplus per worker, as defined in row heading. Ratios in column (4) are estimated by instrumental variables, treating average change in wages of female stayers as the dependent variable, average change in wages of male stayers as the endogenous explanatory variable, and the change in surplus measure as the instrument. Standard errors, clustered by firm, are in parentheses.

- Firm premiums are strongly related to value added per worker, and women capture about 88%–90% of the measured rent-sharing gains captured by men.
- Within-firm wage responses to profitability changes imply almost the same bargaining ratio, reinforcing the rent-sharing interpretation.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that firms are a quantitatively important source of the gender wage gap. Women are both underrepresented at high-premium firms and paid slightly smaller firm-specific premiums than men when employed at comparable firms. Together, these channels explain about one-fifth of the gender wage gap in Portugal.

7.2 Economic intuition

The paper’s intuition is simple but powerful. Firms differ in the rents they share with workers, and labor markets do not assign men and women evenly across those firms. On top of that, the same firm can divide its rents differently across gender. The gender wage gap therefore reflects not only worker-level productivity or preferences, but also the structure of firm wage-setting and who gets access to the best-paying firms.

7.3 Takeaway

This paper is one of the clearest demonstrations that the gender wage gap is partly a firm phenomenon. If one wants to understand or reduce the gap, it is not enough to study workers in isolation. One must also study firm pay policies, worker sorting across employers, and the mechanisms that determine how surplus is shared within firms.